

Privacy-Preserving Federated OCT Retinal Disease Classification using a Hybrid Residual-Attention Network

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Abstract—The field of Optical Coherence Tomography (OCT) has revolutionized ophthalmic disease diagnosis, including Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), and Drusen. Centralized deep learning models are prone to patient data sharing across institutes, thereby creating a huge concern in data privacy, regulation and security. In this paper we introduce a framework for privacy-preserving multi-class OCT retinal disease classification. This framework amalgamates a Hybrid Residual-Attention Network (ResAttnNet) architecture with Federated Learning (FL) and Differential Privacy (DP). Residual learning combined with Squeeze-and-Excitation (SE) attention was utilized for effective feature extraction. FL provided a mechanism to decentralize learning without transmitting any raw data. Formal (ϵ, δ) -privacy guarantees were ensured using calibrated Gaussian noise and gradient clipping to satisfy the requirements of DP. The Non-IID data heterogeneity was simulated using Dirichlet distribution with $\alpha = 0.1$, and Gradient-weighted Class Activation Mapping (Grad-CAM) was added to help for interpretability in the clinical setting. With the proposed framework we were able to achieve 99.27% accuracy in centralized training and 95.4% accuracy in FL with DP at $\sigma = 0.02$, showing a trade-off between privacy and utility.

Index Terms—Optical Coherence Tomography, Federated Learning, Differential Privacy, Deep Learning, Residual Networks, Attention Mechanisms, Explainable AI

I. INTRODUCTION

Retinal disease is one of the world's greatest causes of irreparable vision loss. Diseases including CNV (choroidal neovascularization), DME (diabetic macular edema) and Drusen have the potential to permanently destroy the retina if unnoticed, causing blindness. Clinical standard for visualizing retina has developed to the use of OCT, high resolution imaging capable of resolving structural aberrations at the micro-scale (in a cross-section of retina), and as such makes OCT scans suitable for automated deep learning classification. Despite a success in the classification of medical images, central training in deep learning is often not possible, as medical data often resides on different institutions and regulations such as HIPAA or GDPR are not conducive to central data aggregation. Data breach risks are a problem of "concentration" if only

one dataset exists on a central machine. FL solves this by having institutions locally train on their own data, but share only model updates to a central machine, which aggregates them in a strengthened global model. FedAvg is a popular FL algorithm that achieves this [12]. However, FL alone is not secure from gradient inversion attacks where training images can be reconstructed by someone intercepting updates between clients and server [2], hence DP is needed. DP provides a mathematical guarantee of the amount of privacy leaked per round in a DP (ϵ, δ) definition [9], and DP-SGD injects noise to individual gradients via gradient clipping [5]. The additional difficulty arises from hospitals holding data that follows a Non-IID distribution due to disease prevalence and varying patient demographics which leads to drift in client models in standard FL [7]. Grad-CAM [13] is a technique which produces heatmaps indicating class-discriminative features of model classifications enabling further use in clinical validation. This paper combines these methods in one coherent framework. The main contributions include (1) the ResAttnNet model which combines residual learning with SE attention for the classification of OCT scans, (2) a framework for federated training utilizing FedAvg across simulated hospital clients, (3) the use of Opacus for DP-SGD yielding formal (ϵ, δ) -DP, (4) Non-IID data generation using Dirichlet distributions ($\alpha = 0.1$), and (5) Grad-CAM for class-specific feature interpretability.

II. LITERATURE REVIEW

There has been extensive study on privacy-preserving FL for medical imaging. In [1], Shukla et al. Conducted FL with DP for breast cancer detection with 96.1% accuracy with an $\epsilon = 1.9$ demonstrating feasibility of privacy-utility tradeoffs. Saifullah et al. In [2] showed that DP can severely distort Grad-CAM attribution maps suggesting cautious integration of privacy and explainability for clinical AI. DP-assisted FL system with a robust privacy model is designed and verified for IoMT TB diagnosis that adheres to HIPAA and GDPR

standards by Barnawi et al. In [3]. Xie et al. In [4] integrated both local and centralized DP and presented a two-tiered FL where 92.5% accuracy is obtained and 85% privacy leakage is reduced. An adaptive DP noise based on gradient sensitivity is utilized by Yu et al. In [5] leading to 3.47% improvement in classification accuracy compared to the fixed noise baseline. Shiri et al. In [6] showed valid DP-FL with multi-institutional COVID-19 CT scans that performs comparably to centralized FL and also ensures immunity to membership inference attacks. Khalaf et al. In [7] present hybrid adaptive DP-FL resilient to Non-IID distribution. Adnan et al. In [8] present FL with DP for vehicular context that provides scalable architectural insights applicable to the current setup. Yu and Li [9] introduce dynamic optimization of DP budget with adaptive selection based on sensitivity and outperforming static budget selection. Farooqi et al. [10] provide a survey on DP-FL for IoMT that identify communication overhead and Non-IID data characteristics as the key challenges.

Gu et al. [11] surveyed FL privacy methods for healthcare and acknowledged gradient leakage in FL, reinforcing the motivation of concurrent DP application. Nazir and Kaleem [12] and Sandhu et al. [13] presented surveys on FL for medical imaging applications and identified the problem of Non-IID data as persistent. Rehman et al. [14] pointed out scalability and data heterogeneity are the obstacles to clinical FL implementation. Yang et al. [15] proposed FedDPA which applied layer-wise adaptive DP to fasten the convergence speed. Zheng et al. [16] proposed sensitivity-aware DP in federated medical imaging to enhance accuracy while ensuring the similar privacy budget. Shiri et al. [17] showed federated transfer learning with DP to detect artifact of PET image in across clinic sites. Wassan et al. [18] validated the effectiveness of FL with DP to classify biomedical image, but pointed out the dilemma of privacy-accuracy in Non-IID situation.

To date, none of the existing works have combined FL, DP-SGD, Non-IID simulation and Grad-CAM explainability together for multi-class OCT retinal classification. The present work fills this void.

III. METHODOLOGY

The architecture presented here combines a Hybrid Residual-Attention Network into a differential privacy enhanced and Grad-CAM explainable federated learning system. Figure 1 shows the whole system and Fig. 2 illustrates the federated learning procedure.

A. Hybrid Residual-Attention Network

ResAttnNet utilizes residual blocks together with SE attention module for deep OCT feature extraction with low cost federated friendly architecture.

Residual Learning: Each residual block computes

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \quad (1)$$

where $\mathcal{F}$ is stacked convolutional mapping. Shortcuts with identities allow gradients to not vanish and deeper training without performance loss.

SE Attention: Given feature map $\mathbf{U} \in \mathbb{R}^{H \times W \times C}$, the squeeze step produces a channel descriptor via global average pooling:

$$z_c = \frac{1}{HW} \sum_{i,j} u_c(i,j) \quad (2)$$

The excitation step scales channels by learned importance weights:

$$\mathbf{s} = \sigma(W_2 \cdot \delta(W_1 \cdot \mathbf{z})), \quad \tilde{\mathbf{u}}_c = s_c \cdot \mathbf{u}_c \quad (3)$$

where δ is ReLU and σ is sigmoid. This recalibration enhance the diagnostic relevant channels, meanwhile suppressing noise to make it easier to discriminate lesion specific signatures like the subretinal fluid(CNV) and drusen deposits. This network finally comprised with several residual-SE modules, global average pooling, a full connected layer and softmax over the four categories. Dropout is utilized in front of the classifier to prevent overfitting.

B. Federated Learning with Non-IID Simulation

For every round of communication t , the server sends global parameters w^t to all K clients. The client k trains locally on $\mathcal{D}_k$ and sends its parameter update w_k^{t+1} back. The server aggregates with FedAvg [1]:

$$w^{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{t+1} \quad (4)$$

where $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$. Original images will never be sent. To provide real-world hospital heterogeneity, partitioning data is done via the Dirichlet distribution [7]:

$$p_k \sim \text{Dir}(\alpha), \quad \alpha = 0.1 \quad (5)$$

(Table I). The global model is initialized using a warm start from centralized pre-trained weights to achieve faster convergence on Non-IID clients.

C. Differential Privacy

We do not share raw data in FL. However, the gradient updates are vulnerable to inversion attacks [2]. We perform DP-SGD using Opacus [5] in each client (Fig. 3). We assume that algorithm $\mathcal{M}$ satisfies (ϵ, δ) -DP if: For neighboring datasets D, D' . Gradients are clipped at the sample level:

$$\Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{S}] + \delta \quad (6)$$

For neighboring datasets D, D' . Gradients are clipped at the sample level:

$$\bar{g}_i \leftarrow g_i \cdot \min\left(1, \frac{C}{\|g_i\|_2}\right), \quad C = 1.0 \quad (7)$$

then perturbed with Gaussian noise:

$$\tilde{g} = \frac{1}{B} \sum_{i \in \mathcal{B}} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \quad (8)$$

And $\delta = 10^{-5}$ and $\sigma \in \{0.02, 0.05\}$ controls the privacy-utility trade-off. The accumulation of ϵ is recorded through Rnyi DP accountant by Opacus.

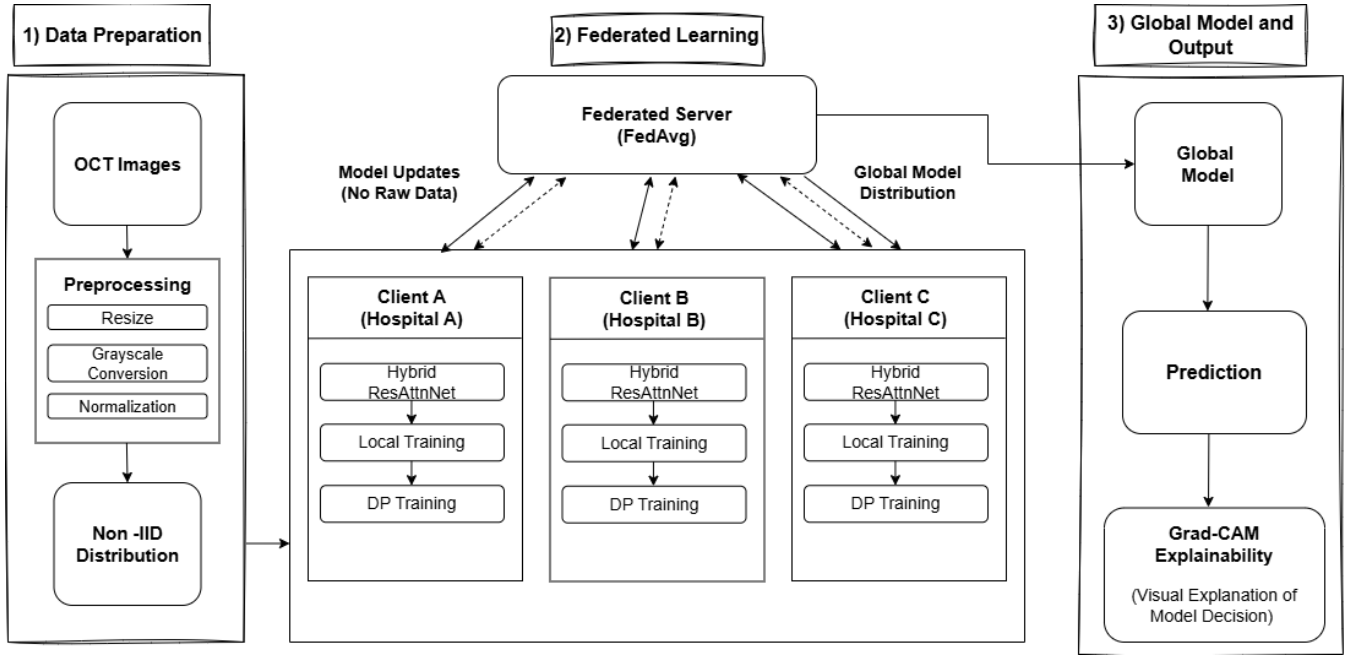


Fig. 1. Proposed system architecture: end-to-end federated learning pipeline with differential privacy and Grad-CAM explainability across three simulated hospital clients.

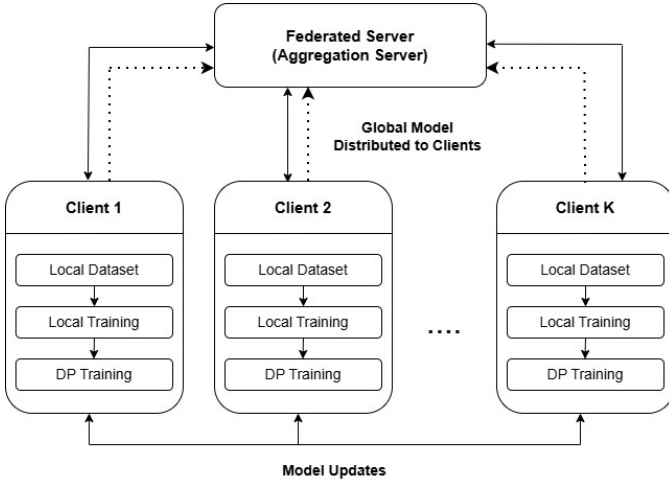


Fig. 2. Federated learning workflow: global model distribution, local DP-SGD training, and FedAvg aggregation.

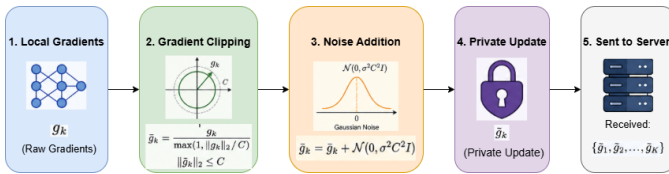


Fig. 3. Differential privacy mechanism: per-sample gradient clipping followed by Gaussian noise injection before server transmission.

D. Explainability via Grad-CAM

Using the same final-layer feature maps A^k , Grad-CAM [13] creates class discriminative heatmap $L_c^{\text{Grad-CAM}}$ by weigh-

ing feature maps using scores from gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{ij}^k}, \quad L_c^{\text{Grad-CAM}} = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (9)$$

Upsample heatmap to the resolution of the input image, and overlay the result over the OCT scan. The procedure is not invasive-the model weights do not change-and it's compatible with the federated setup.

IV. EXPERIMENTAL SETUP

A. Dataset and Preprocessing

All experiments applied use publicly available OCT2017 Retinal Image Dataset (Kermany et al.). There are four classes: CNV, DME, DRUSEN and NORMAL. Fig. 4 shows an example image. Images were resize to 224×224 pixels, convert the 1-channel images to 3-channel format, and normalized per-channel. Images were randomly horizontally flipped and rotated in order to perform data augmentation. The dataset was distributed among 3 clients using non-IID method based on Dirichlet sampling ($\alpha = 0.1$); see Table I for the distributions of the four classes.

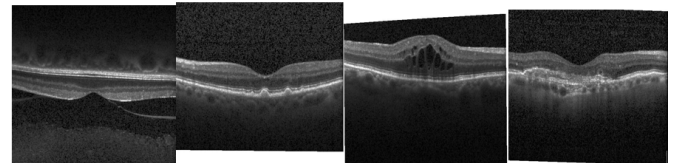


Fig. 4. Representative OCT images: (a) CNV, (b) DME, (c) DRUSEN, (d) NORMAL.

TABLE I
NON-IID CLASS DISTRIBUTION ACROSS CLIENTS (DIRICHLET $\alpha = 0.1$)

Client	CNV	DME	DRUSEN	NORMAL
Client A (CNV-biased)	71%	11%	9%	9%
Client B (DME-biased)	8%	74%	10%	8%
Client C (DRUSEN-biased)	7%	9%	75%	9%

B. Training and FL Configuration

ResAttnNet was implemented using PyTorch, trained on NVIDIA GPU. The Adam optimization algorithm was used, with learning rate 1×10^{-4} and batch size 32. The FL was run for 3 communication rounds with 1 local epoch per client for each round, utilizing FedAvg, over 3 clients (Hospital A, B, C). Gradient clipping norm $C = 1.0$ was applied with $\delta = 10^{-5}$. The two DP configurations used were moderate privacy, higher accuracy configuration, $\sigma = 0.02$ and strong privacy, lower accuracy configuration, $\sigma = 0.05$. The system implementation consisted of a FastAPI back end, and coordinator interface using React, which supported real time FL monitoring and Grad-CAM rendering.

C. Evaluation Metrics

Overall we compute the classification accuracy and per-class AUC-ROC for all configurations. As a qualitative measure of the clinical relevance of learned features, we utilize Grad-CAM visualizations.

V. RESULTS AND DISCUSSION

Table II summarizes model performance across all experimental configurations.

TABLE II
MODEL PERFORMANCE UNDER DIFFERENT TRAINING SETTINGS

Method	Noise (σ)	Accuracy (%)
Centralized Learning	0	99.27
Federated Learning (IID)	0	99.38
FL + DP (Non-IID)	0.02	95.40
FL + DP (Non-IID)	0.05	81.90

A. Centralized and Federated Performance

The accuracy achieved using ResAttnNet under centralized training is 99.27%, establishing the upper-bound performance benchmark. From the multi-class ROC curves in Fig. 7, we observe robust per-class discriminability, obtaining AUCs of 0.9990 (CNV), 0.9952 (DME), 0.9999 (DRUSEN), and 0.9996 (NORMAL). Looking at the confusion matrix shown in Fig. 6, the classification is well-balanced and exhibits low inter-class confusion, particularly in the closely similar DME and DRUSEN classes, where the SE channel attention has focused on the relevant features for diagnosis.

Under the IID federated setting without DP, the global model yields 99.38%, which is merely a slight decrease of less than 0.4% compared to the centralized one. It can be seen that FedAvg is able to effectively aggregate local models when

their local data distributions are the same. The training process was consistent and never diverged across all communication rounds.

B. Privacy-Accuracy Trade-off

The privacy-utility trade-off curve is presented in Fig. 5. With $\sigma = 0.02$ and Non-IID setup, FL+DP achieves 95.4% accuracy, which is 3.9% less than centralized. This confirms the possibility to apply non-prohibitive accuracy degradation to achieve formal DP guarantees as shown in [5]. When $\sigma = 0.05$, the accuracy degrades to 81.9%, which is attributed to increased noise level causing gradient signal distortion and affecting convergence. It is suggested to use $\sigma = 0.02$ as the operational point for clinical applications. FedAvg handles client model drift by weighted aggregation, while central weights initialized for warm start in [7] improves convergence speed.

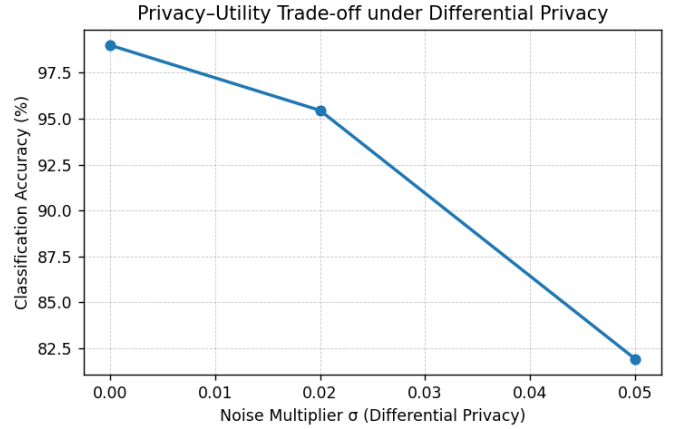


Fig. 5. Privacy-utility trade-off: classification accuracy vs. noise multiplier σ .

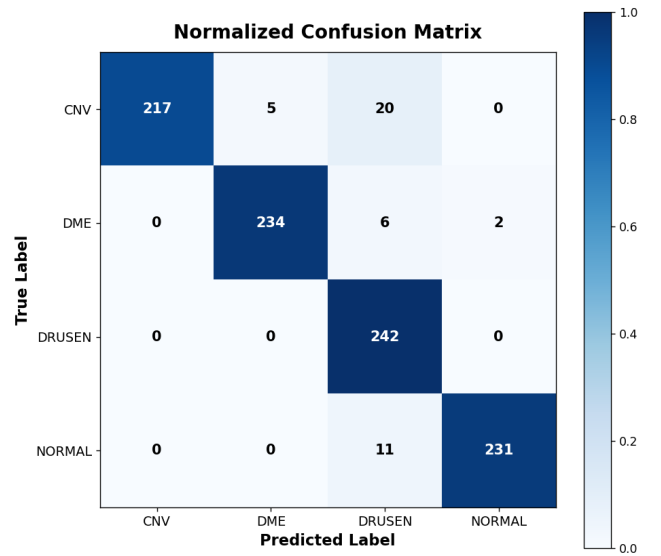


Fig. 6. Confusion matrix for FL + DP ($\sigma = 0.02$).

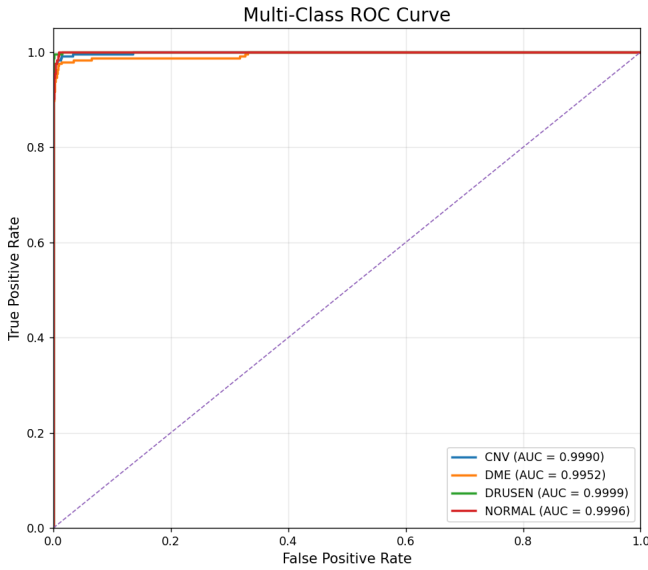


Fig. 7. Multi-class ROC curves. AUC: CNV=0.9990, DME=0.9952, DRUSEN=0.9999, NORMAL=0.9996.

C. Grad-CAM Qualitative Analysis

Figure 8 shows the Grad-CAM heatmaps of each class. In CNV, the attention is focused on the choroidal neovascular membrane in the RPE layer or below. In DME, the heatmaps focus on the cystoid fluid areas inside the inner layers of the retina. In DRUSEN, the heatmaps are distributed around the RPE-Bruchs membrane boundary. In NORMAL class, the heatmaps are dispersed with weak attentions and shows that there are no local pathologies in the image. We can observe that the region of attention spatially agrees with the clinical loci of the diseases, which gives an indirect and qualitative proof of the diagnosable meaningfulness of the learned features, instead of accidental spurious correlation [2].

VI. CONCLUSION

This paper proposed a privacy-preserving FL system to conduct automatic multi-class OCT retinal disease classification. With the proposed residual learning combined with SE attention module, ResAttnNet reached 99.27% and 99.38% on centralized and FL IID condition respectively (negligible FL overhead for homogeneous data), while at Non-IID condition, with DP-SGD at $\sigma = 0.02$, accuracy could reach 95.4% under (ϵ, δ) -DP guarantee (controllable and measurable privacy-utility trade-off). Furthermore, Grad-CAM visualizations for all the four diseases demonstrated clinically relevant feature maps of lesion localization, supporting diagnostic explainability. Limitations of this work include, using a simulation of three-client FL settings instead of three hospitals or multiple healthcare institutions, applying fixed DP noise rather than dynamically adjustment of σ per round, and no formal validation by ophthalmologists regarding the relevance of Grad-CAM interpretation. For future work, we aim at realizing large-

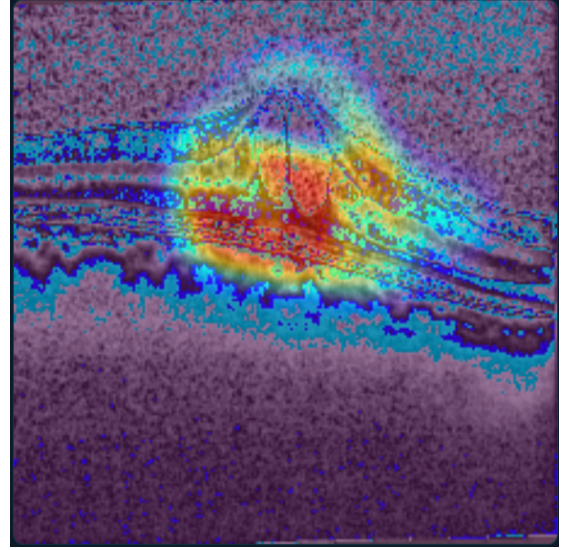


Fig. 8. Grad-CAM saliency maps for CNV, DME, DRUSEN, and NORMAL, highlighting clinically relevant retinal regions.

scale multi-hospital FL with actual clinical data. It should incorporate dynamically adjusted DP noise and integrate with PACS and EMR systems for deployment [18] or use sensitivity aware σ adjustment per round [16].

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